

FIGURE · ALGORITHM 1

FIP-PPFL System Initialization and Reference Bootstrapping

A geometric view of the three setup phases — key generation & registration, model initialization, and reference bootstrapping — culminating in a unit-norm trusted reference direction $g_{\text{ref}}^{(o)} \in \mathbb{R}^d$ stored at the server-side FIP Verifier module for round 1.

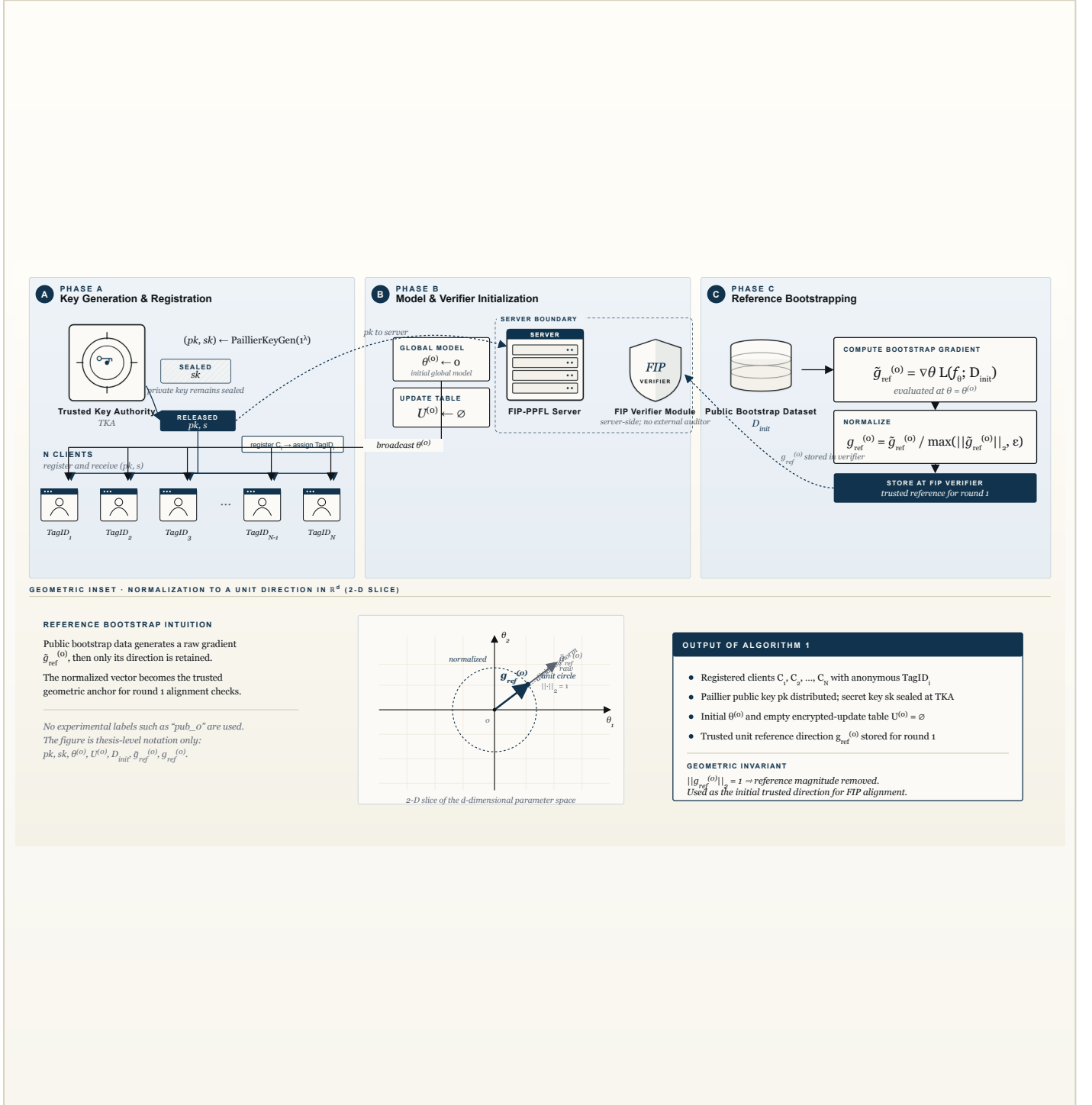


Figure 1. Algorithm 1. The TKA generates the Paillier key pair and keeps sk sealed while releasing only pk and the fixed-point scale S . The server initializes $\theta^{(o)} \leftarrow \text{InitModel}(f_\theta)$ (zero initialization for the logistic-regression experiments) together with $U^{(o)} = \emptyset$, broadcasts $\theta^{(o)}$ to the registered clients, and the server-side FIP Verifier stores the initial trusted reference direction. The trusted public bootstrap set D_{init} produces the raw gradient $\tilde{g}_{\text{ref}}^{(o)}$, which is normalized to $g_{\text{ref}}^{(o)}$ using $\max(\|\tilde{g}_{\text{ref}}^{(o)}\|_2, \epsilon)$. The bottom inset shows the geometric normalization from a raw gradient vector to a unit direction in a 2-D slice of the parameter space.